The Fuzzy Logic Controller Used in the Moonrat II

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1 Fuzzy Logic Controller Strategy

Precise and adaptive temperature control is essential in cultivation chambers for Petrifilm samples. To achieve this, we implemented a *Mamdani–type fuzzy controller*. This approach is efficient and easy to apply to systems with slow dynamic evolution, such as temperature and heat control processes [1, 6, 4, 3, 5].

To enable integration into an embedded system, we employed the *Embedded Fuzzy Logic Library (eFLL)* developed by AJ Alves [2]. This library provides a standardized implementation of fuzzy control for embedded computing platforms.

The system relies solely on measuring the chamber temperature and a predefined reference value. Consequently, we designed a *Mamdani–based fuzzy control scheme* with two input variables:

- **Error** (e): The difference between the measured temperature $temp_c$ and the desired temperature $temp_r$:

$$e = temp_c - temp_r \tag{1}$$

- **Error difference** (Δe): The discrete change in error between two sampling instants:

$$\Delta e = e(t_k) - e(t_k - 1) \tag{2}$$

We defined both input variables using three fuzzy sets: *negative, zero, and positive*. We established the relationship between these variables and the controller's response through a *set of fuzzy rules*, as detailed in Table 1.

The system's output variable is the *PWM frequency*, which regulates the power supplied to the thermal pad responsible for heating the chamber. We defined this output using three fuzzy sets: *high, medium, and low frequency*.

We carried out the fuzzy system design, including the membership functions of the input and output variables, using the *Fuzzy Logic Designer* toolbox in MATLAB. Figure 1 illustrates the configuration of the membership functions and the assigned ranges for each fuzzy set.

2*Control		Error Difference		
		Negative	**Zero**	**Positive**
3***Error**	**Negative**	P	P	Z
	Zero	P	Z	N
	Positive	Z	N	N

Table 1: Fuzzy control rule table. P = Positive, Z = Zero, N = Negative.

Finally, we defined the *defuzzification process* and the configuration of membership degrees according to the parameters shown in Figure 2.

References

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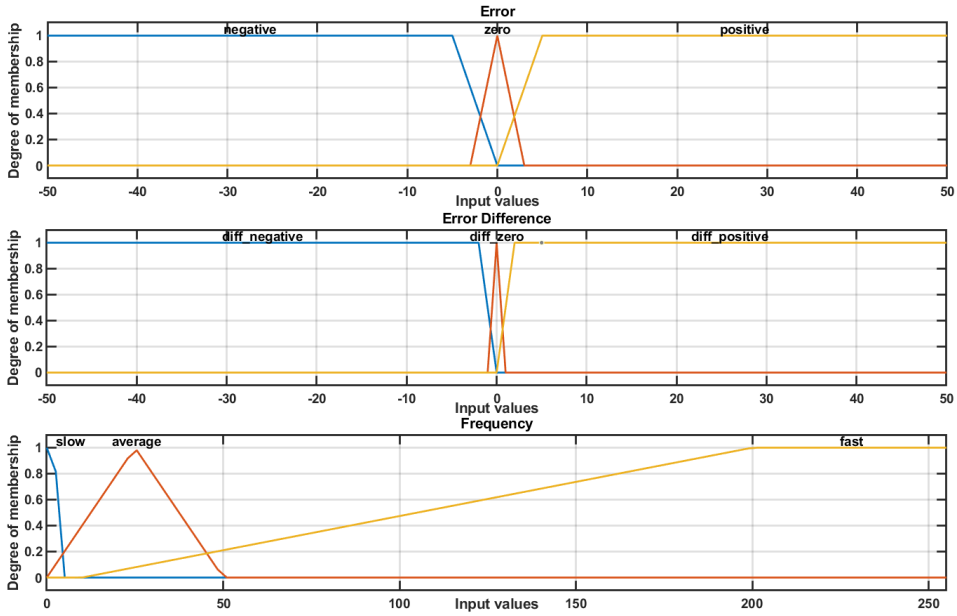

Figure 1: Membership functions for the fuzzy system inputs and outputs, designed using the Fuzzy Logic Designer toolbox in MATLAB.

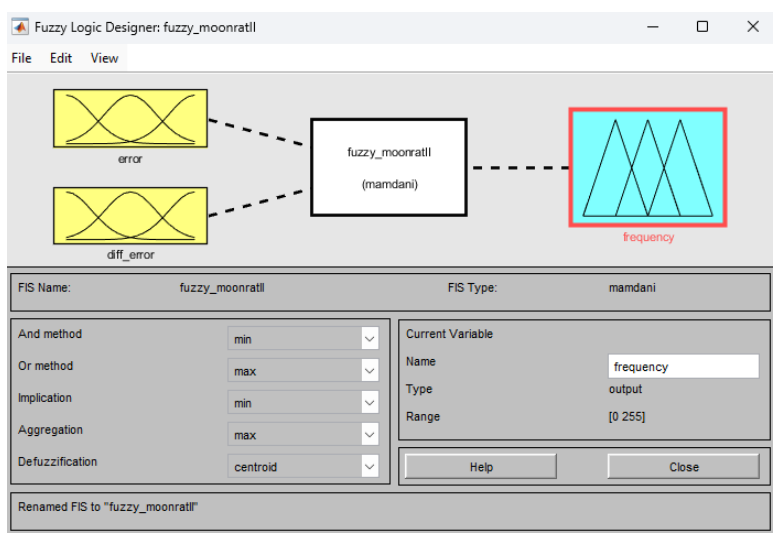

Figure 2: Defuzzification and membership function configuration for the fuzzy control system.